

Exploring Secondary Teachers' Needs and Values in Culturally Responsive Teaching

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ABSTRACT

Culturally Responsive Teaching (CRT) is educational practices that acknowledge, celebrate, and incorporate students' identities and backgrounds into the classroom. Technologies can be used to ease the challenges teachers face with implementing CRT-related practices including inadequate guidance and knowledge of students' cultures. Value sensitive design (VSD) accounts for human values in the design of technology. VSD can be used in the design of technological systems for helping CRT-related practices. However, the CRT context poses unique challenges due to the interactions between teachers' values and other socio-material factors such as students' values, community characteristics, and available resources. This preliminary study uses a low-fidelity cultural tool to elicit teachers' needs and values on CRT from various cultural backgrounds and educational settings. The results of the interview analysis provide an initial understanding of how needs, values, socio-material factors, and educational contexts are intertwined, and how they can be used to inform the design of technologies that assist teachers with CRT.

CCS CONCEPTS

• Human-centered computing → User studies.

KEYWORDS

Education/Learning, Secondary Teachers, Culturally Responsive Teaching, Value Sensitive Design

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1 INTRODUCTION

Culturally Responsive Teaching (CRT) denotes a set of instructional practices that incorporate students' cultural identities into educational materials and activities for effective and equitable learning experiences [12]. Teachers face various challenges in implementing CRT-related practices such as insufficient knowledge of students' cultures and a lack of training and resources [5, 19, 23]. To

help with these challenges, technology can be designed to assist teachers with their CRT-related practices such as increasing their cultural knowledge. In designing such technologies, a Value Sensitive Design (VSD) approach can provide effective guidelines and frameworks [9]. Through VSD, designers can understand and incorporate users' values in designing technological systems that are well-perceived and meet their intended goals [7, 10]. However, the context of CRT poses unique challenges in understanding users' values. The CRT context presents entanglements of students' cultural traits and teachers' values that are often moderated by resource constraints, community characteristics, and policies [14, 23, 26, 30]. As an initial step of a nuanced VSD approach, as indicated by Le Dantec and colleagues [20], it is necessary to closely examine teachers' CRT-related practices in different geographical, socioeconomic, and school contexts to help inform the design of technologies that support teachers' inclusive teaching in the class environment. While it is necessary to examine both teachers and students to fully understand the landscapes of the values embedded in CRT, this poster focuses on teachers first, because identifying their current practices and perception of students' values are a precursor to examining students' values and their interactions with teachers' CRT. In this poster, we report preliminary findings from interviews with secondary school teachers in the United States (grades 6-12/student ages 11-18). This study aims to understand teachers' needs and values in CRT in relation to other socio-material factors and contexts through leveraging a low-fidelity tool, used as a technology probe [18], which provides cultural data for a school district.

2 COMPLEXITY OF THE CRT CONTEXT

One of the main goals of CRT is to place importance on the diverse cultural backgrounds of students to make them feel seen in the classroom (e.g., acknowledgment and a sense of inclusion) and increase their engagement [12]. Through a qualitative synthesis, Piazza et al. [23] uncovered five strategies used to teach diverse learners in terms of language and culture: dialogue, collaboration, visual representation, explicit instruction, and inquiry. Despite the identified solutions, teachers often encounter many challenges when implementing these strategies. Causes of such difficulties include a lack of understanding of their students' cultural identities [5, 8, 23] and intercultural education [13].

VSD and related approaches have been successfully used to increase the effectiveness of CRT. CRT-related research that takes the VSD approach includes designing students' and instructors' values concerning mobile education and usability [28] and considering cultural backgrounds as the main focus in understanding values in the design of their systems [1, 22]. Outside of VSD-based approaches, the broader HCI community has focused on increasing people's

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Table 1: Participant Demographics

ID	Gender	Race	Years of Experience	Subject	Grade	Socioeconomic Status	Racial Diversity	Area Type	State
P01	F	White	8	Art	9-12	mixture	low	rural	IA
P02	M	White	22	Biology, Forensic Sci.	9-12	mid-low	high	urban	RI
P03	F	White	4	Science	6	low and high	low	urban	MD
P04	F	White	21	English Lang. Arts	7-8	mixture	high	urban	CT
P05	F	Hispanic	4	English	9	low	moderate	urban	AZ

cultural awareness in non-educational contexts [3, 4, 6, 22]. In Education, researchers have developed CRT-assisting technologies, mainly as tools to be used by students [11, 16, 17, 24, 32].

Though these studies provide a basis to designing technology for CRT, there is a gap in the literature. While the CSCW and HCI literature has focused on the examination of users' values, they paid less attention to how their values are formed through interacting with socio-material factors surrounding them in the educational contexts. In Education, meanwhile, CRT-focused studies have concentrated on the student-side technologies, but have not considered teachers' values and needs closely in designing technologies. This suggests that designing technology for CRT may require a broader understanding of the context by focusing on the interactions between stakeholders' values and surrounding socio-material factors such as students' values, community characteristics, and available resources. This is because teachers' educational practices may manifest differently depending on their students' values, community characteristics, and available resources, just as they do with technologies [2]. To fill the identified gaps, we aim to examine CRT-related needs and values of secondary teachers from various cultural backgrounds and settings that were brought out through their interaction with a low-fidelity prototype of our new cultural tool.

3 METHODOLOGY

3.1 Data Collection

We conducted unstructured interviews, via Zoom [34], with five U.S. secondary public school teachers between Dec 2021 and Feb 2022. We recruited them through publicly available school websites. The interviewees varied by location, subject area, grade level, years of experience, school's socioeconomic status, and racial diversity.¹ We list the demographics of the participants in Table 1. Educational contexts that may have affected teachers' CRT practices are observed: the teacher's years of teaching experience, subject area (i.e., STEM² vs. non-STEM), the school's grade level, the school district's socioeconomic status, and the school district's diversity level.

During interviews, we asked participants to think aloud while they explored a design mock-up of a low-fidelity cultural tool [29] that included various visualizations of cultural data on a fictional school district (Figure 1). This tool was used as a means to elicit teachers' needs, values, and challenges. There were no predefined interview questions; instead we asked participants to express their thoughts aloud while they explored our tool. When fitting, we asked

follow-up questions to get further details. The tool included data on the community's demographics, households, schools, hobbies, places to visit, the option to compare school districts, and recommended CRT-related practices based on the data. The selection of data and their visualizations were informed by findings from our previous study that explored teachers' use of culture in the classroom [30].

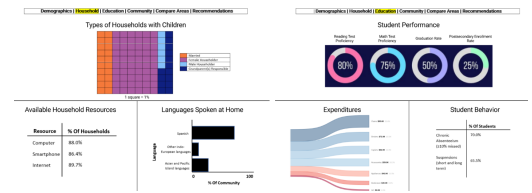


Figure 1: Screenshots of two tabs of the design mock-up: household data (left) and education data (right) for a hypothetical district.

3.2 Data Analysis

We transcribed each of the interviews and analyzed it using a grounded theory approach [31]. We started with open coding [27] which led to the development of a codebook about their values, needs, educational contexts, and any interactions between them. After our initial coding that provided a descriptive overview of the data [21], we revisited the data and applied value coding where appropriate. To identify participants' values, we looked for phrases like "I think," "I feel," "I want," "It's important that," "I like," "I love," and "I need" [27]. After value coding, we used constant comparative analysis [15] to form categories/themes. Through iterative refinements of codes [33], we identified the following five dimensions of teachers' needs and values in implementing CRT: (i) actionable items, (ii) community connections, (iii) transparency, (iv) social awareness, and (v) knowledge of neighborhood and institution.

4 TEACHERS' NEEDS AND VALUES IN CRT

4.1 Values

4.1.1 Community Connections and Social Awareness. Participants expressed the importance of connecting with their community both in and out of school. Some participants believed that learning about the activities that are accessible in their community would be helpful for getting their students involved in after-school activities. Sometimes students are not aware of what they could do outside of school, or they show interest in a culturally-tailored activity, but it is not available to them in school. So, participants P02 and

¹Participants self-identified their race as well as the diversity level and socioeconomic status of the school.

²STEM represents Science, Technology, Engineering, and Mathematics.

P03 imagined that having a list of community places and activities to share in the classroom would help with that issue. In addition, participants felt that information about local activities would be useful for connecting their lessons with the community, primarily through field trips. For example, P01 who teaches a non-STEM subject, noted how she was always trying to think of ways to relate the curriculum back to the students' community so that they can connect more with the lesson: *"What are the places that we have available to us that we can get out and make their learning more hands on and relevant to them?"*

To make further community connections, specifically with the students, participants were concerned with being cognizant of the people in their community and their lifestyles, or in other words their social awareness [25]. Participants who teach in high school were interested in knowing the religious makeup of their community because it would give them an informed idea of their students' religion. P02 felt it could help him be sensitive when teaching Evolution in Biology by reaching out to students' families for guidance. P01 noted that knowing students' religion would help teachers cultivate students' social awareness by exposing them to religions outside of theirs using religious artwork.

Outside of religion, students' home life including the types of households and their available resources (Figure 1) seemed of value to participants. Specifically, we found that participants who teach students from a lower socioeconomic household believed that understanding students' home lives would be helpful in increasing their own social awareness since, oftentimes, the students' situation might be very different from their own. When viewing the household data, P02 discussed how having such data would improve his understanding of and interactions with his students: *"What I frequently have trouble remembering is [...] that my students have a whole life outside of school, and [...] is very likely very different from my life outside of school [...]. Having real numbers to be able to connect to and real reliable data to connect to I think would make that easier for me."*

4.1.2 Transparency. When showing a hypothetical graph that would provide a school district's expenditures (Figure 1), the value of transparency emerged. The participants who highlighted the importance of transparency were those with the least amount of teaching experience (P03 and P05). Both participants felt that including the expenditures was very important because it would enable all teachers to openly see how the district allocated funds. P03 said *"I just think that's like transparency data about what resources each district has"* and P05 felt that the data made things open and she was happy to *"see where they money's going"*. Oppositely, the participants with the most amount of teaching experience (P02 and P04) did not mention anything about transparency and instead felt like the expenditure data, as discussed, would not be useful to them. P02 wished to compare his and another district's expenditures, while P04 wanted to know how much is spent on *"technology, student computers, teacher salaries, and art"* instead of the included categories so she could better meet her students' needs.

4.2 Needs

4.2.1 Knowledge of Neighborhood and Institution. Participants value being socially aware of their community and using that

information to connect with the community. This leads to our discovery of participants wanting to have knowledge of their school and its surrounding neighborhood. This helps them have a more focused view of the community, the local residents, and their activities. This need was particularly seen in high school teachers. They believe that the school district would not be a good representation of the students they teach, and would prefer school specific data; as P05 states *"School level is all about neighborhood. It's more catered to where you're at."* Participants also brought up how there could be certain schools in the district that influenced the data, both positively and negatively, and thus may misrepresent the participant's school. P01 mentioned *"depending on where you're at in the school district, that student might be living a whole different experience."*

In addition, participants were eager to know about their specific school so they can pinpoint their school's issues or things to address. When viewing the possible demographic and educational data, P02 shared how he was aware that his school differed demographically and academically from other schools, but did not know why. His values were on having the data about schools that helps explain these differences and recommend any specific actions to help his students. He said *"If I know the facts about my school and no one can say to me well that's because this is mixed in. That's nice to know—this is exactly what's going on in our school. What can we do to fix it?"*

4.2.2 Actionable Items. The desire to acquire knowledge of one's own school and neighborhood gives rise to teachers' needs for actionable items. Once teachers have this knowledge, they want to be provided with tasks they can do to incorporate it in CRT-related practices. Teachers value being able to navigate actions that can help respect and acknowledge students' cultural identities. Some participants mentioned how having specific recommendations for actions in the tool would be valuable. P01 who teaches at a predominantly white school said she was often reminded of CRT, but was not able to implement it, because she did not know if she would be offending them. To help with this, P01 said she would want to see particular actions to do, like *"do not assign online homework"*, and then an explanation for why. She said that *"We get reminded of [CRT] frequently, but it's not something that we have a ton of tools and understanding for [...]. How am I building if the way I teach is not culturally adequate? [...] then I'm getting frustrated that it's not working."* P03, who teaches at a school with moderate diversity, recognized that, because she was "middle-class and white," she might not identify or relate to some students outside of that demographic group such as students from low-income families or those who immigrated from a war-torn country. Because of this, she showed an interest to have some recommendation on books that could help her become more culturally responsive.

5 DISCUSSION AND FUTURE WORK

Through the preliminary exploration of teachers' needs and values on CRT practices, we were able to uncover challenges, tensions, and values in various contexts. The findings provide implications for the design of CRT-supporting systems. First, the findings suggest that teachers from schools with low diversity need more direction when interacting with students from diverse backgrounds, because of their infrequent interaction with students from different cultural

backgrounds than their own. CRT designers can benefit from understanding the interaction between teachers' values and school diversity by recommending specific CRT actions teachers can take depending on a school's diversity instead of general guidance.

Second, the interaction between teachers' subject type and the community characteristics might shape an important decision point for designers. Participants from non-STEM subjects valued connecting their lessons with the community. Among them, the participant in a rural area located relevant activities easier than those in urban areas. While urban areas typically have more cultural activities available, these activities may be harder to locate than in rural areas. In other words, teachers' values on their subject areas may manifest as CRT-related practices by interacting with available resources, activities, and opportunities that come from the urbanness of the community that the school belongs to. The identification of such interactions can help CRT designers create and adapt systems to assist teachers in different subject areas and urban settings with exploring resources and opportunities that are available in their community.

Finally, the school's grade level may give rise to the shaping of teachers' values on social awareness. Participants teaching high school valued social awareness of their neighborhood and institution more than middle school teachers do. High school teachers interact with older students, thus they may feel an increased responsibility to expose them to real-world lessons outside of school, so that students can prepare to become adults. These findings suggest that developing CRT practices for teachers may have to consider the fact that teachers' needs, values, and concerns are not consistent across grade levels. With the enhanced understanding of the roles of grade level, designers can parameterize users' needs for social awareness that may change based on the students and communities they interact with.

Future studies should explore other socio-material factors such as students' values by triangulating them with teachers' values to see if any tensions exist. Through further interviews with teachers in broader contexts and the theorizing of the interactions between socio-material factors, we hope to gain more insight on the stakeholders' value structures and their CRT-related practices, so that we can inform the design of CRT-supporting systems.

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